Aluminium and other elements in selected herbal tea plant species and their infusions.

The determination of Al, B, Cu, Fe, Mn, Ni, P, Zn and Ca, K, Mg by inductively coupled plasma optical emission spectrometry (ICP-OES) and flame atomic absorption spectroscopy (FAAS), respectively, in digests and infusions of Hibiscus sabdariffa (petals), Rosa canina (receptacles), Ginkgo biloba (leaves), Cymbopogon citratus (leaves), Aloe vera (leaves) and Panax ginseng (roots) was carried out in this study. Particular attention has been given to Al and heavy metals for the identification of possible raw material contaminants, their transformation into the infusion and for predicting their eventual role in the human diet during daily consumption. Additionally, Ion Chromatography (IC) speciation of Al in the leachates was carried out. In dry herbs, hibiscus and ginkgo appeared to contain the greatest contents of Al, Fe, K, Mn, Ni, Zn and B, Mg, P, respectively. A. vera contained the highest amount of Ca and highest values of Cu and P were observed in ginseng. In infusions, the topmost concentrations of Al, B, Cu, Fe, P, K, Mn, Ni, Zn were detected in those prepared from hibiscus petals, Ca from aloe leaves and Mg from leaves of ginkgo. According to a possible daily consumption exceeding 1 L, hibiscus decoction was identified as potentially dietetically significant in the content of certain elements. It seems to be possibly one of the top contributors of B from food (up to 5.5±0.2 mg/L). The Mg contained in the infusion (up to 106±5 mg/L) may be a contributor in the attenuation of blood pressure. A high amount of accessible Mn (up to 17.4±1.1 mg/L) can probably have an adverse effect in humans. The total Al allowance (up to 1.2±0.1 mg/L) suggests that no more than 1 L of the hibiscus infusion should be consumed per day by sensitive individuals including pregnant women and should be completely excluded from the diet of children under 6 months of age and children with chronic renal failure.